CoRoT-2b

R-0097

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_CoRoT-2_180511` · created 2026-07-16T02:05:17+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458249.9126 +/- 0.0022 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.176 +/- 0.029
Transit depth (Rp/Rs)^2	3.09 +/- 1.03 [%]
Orbital Inclination (inc)	85.38 +/- 1.97
Airmass coefficient 1 (a1)	1.012 +/- 0.018
Airmass coefficient 2 (a2)	-0.009 +/- 0.017
Scatter in the residuals of the lightcurve fit is	1.82 %
Best Comparison Star	#2 - [103, 166]
Optimal Aperture	4.13
Optimal Annulus	14.33
Transit Duration (day)	0.084 +/- 0.013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.176 ± 0.029 vs 0.1667 (deviation 0.32σ)
Duration fit vs published	0.084 d vs 0.095 d (deviation 0.85σ)
Mid-time O–C	-4.3 min
Depth SNR	6.1
Residual scatter	1.82 %

Light curve

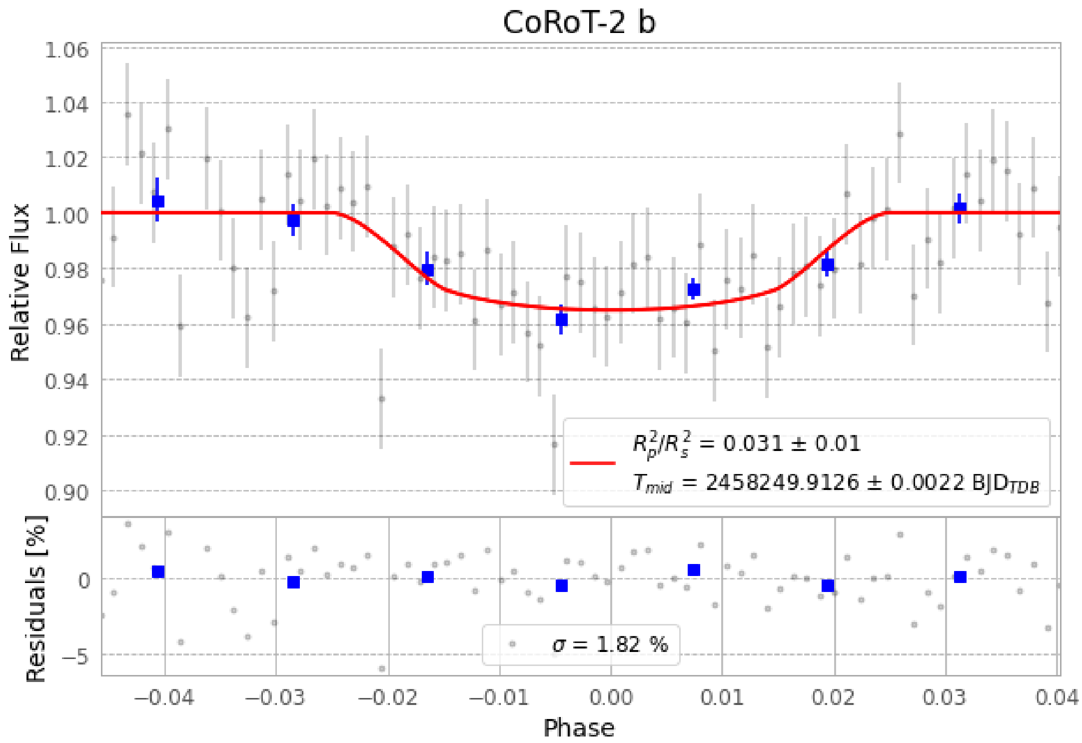

Figure 1 — FinalLightCurve_CoRoT-2 b_2018-05-11.png (SHA-256 07b6cc0b8e405339…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [362, 163], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 8958cc514e429e39…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13`

Data provenance

78 FITS frames (51 MB), `CoRoT-2180511073912.FITS` → `CoRoT-2180511113012.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-180511.log (6664b3f6bf1e…), FinalLightCurve_CoRoT-2 b_2018-05-11.png (07b6cc0b8e40…), FinalLightCurve_CoRoT-2 b_2018-05-11.csv (cb50ab6d3547…)

Compute resources

Wall time 215.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-07-16 02:05Z.